The Rand Index and the Adjusted Rand Index, from the ground up

This note builds RI and ARI from first principles. No leaps, no "it can be shown". Every quantity is computed on small examples that you can verify by hand. By the end you will see that ARI is not a mysterious formula — it is the unique answer to one very specific question, and you will know which question.

1 The setup: comparing two groupings of the same items

We have N items (in our project: articles). Two different "graders" have each sorted them into clusters.

- The **truth** T: the real authorship labels. Truth partitions the items into clusters $T_1, T_2, \ldots, T_K$ of sizes $\alpha_1, \alpha_2, \ldots, \alpha_K$.
- The **prediction** P: what K-means produced. Prediction partitions the items into clusters $P_1, P_2, \ldots, P_L$ of sizes $\beta_1, \beta_2, \ldots, \beta_L$.

We want a single number that measures how well P matches T. Higher should mean better. That is the only goal of this entire document.

The first thing that goes wrong. The obvious idea — "count how many items have the same label in T and P" — does not work, because the cluster labels are arbitrary names. If truth calls a cluster "A" and the prediction calls the same cluster "7", the prediction is perfect even though no item shares a name with the truth. And it gets worse: K-means might split true cluster A into two predicted clusters, or merge two true clusters into one — there is no clean re-labelling that fixes this.

2 The trick: don't compare labels. Compare relationships.

Here is the move that makes everything work.

Forget about cluster names entirely. For every *pair* of items (i, j), T makes one of two statements:

- "i and j are in the **same** true cluster", or
- "i and j are in **different** true clusters".

P makes one of the same two statements. These statements do not depend on what the clusters are called. They only depend on the *partition*, i.e. the grouping itself.

So we ask one question, repeated over every pair of items:

> *Do T and P make the same statement about this pair?*

If we just count how often the answer is yes, we get a similarity score that is immune to label-renaming. That count, divided by the total number of pairs, is the Rand Index.

3 The four kinds of pairs

For each pair (i, j) there are four possibilities, depending on what T says and what P says. We give each combination a letter — this is the notation everyone uses.

	Pred says SAME	**Pred says DIFF**
Truth says SAME	a: both say together (true positive)	b: pred wrongly split a true cluster
Truth says DIFF	c: pred wrongly merged two true clusters	d: both say apart (true negative)

The shaded cells a and d are the **agreements**; the unshaded cells b and c are the **disagreements**. Every pair of items lands in exactly one of the four cells, so

$$a + b + c + d \;=\; \binom{N}{2}.$$

4 A small worked example

Let us pick a tiny example and stick with it for the rest of the document.

Take $N = 6$ items labelled $1, 2, 3, 4, 5, 6$. Truth and prediction:

$$T \;=\; \{\{1,2,3\},\ \{4,5,6\}\}, \qquad P \;=\; \{\{1,2,4\},\ \{3,5,6\}\}.$$

So truth groups the first three together and the last three together; the prediction is the same except items 3 and 4 are swapped.

Total pairs: $\binom{6}{2} = 15$. Let us list every single one.

Pair	In same true cluster?	In same pred cluster?	Category
$(1,2)$	yes (T_1)	yes (P_1)	a
$(1,3)$	yes (T_1)	no	b
$(1,4)$	no	yes (P_1)	c
$(1,5)$	no	no	d
$(1,6)$	no	no	d
$(2,3)$	yes (T_1)	no	b
$(2,4)$	no	yes (P_1)	c
$(2,5)$	no	no	d
$(2,6)$	no	no	d
$(3,4)$	no	no	d
$(3,5)$	no	yes (P_2)	c
$(3,6)$	no	yes (P_2)	c
$(4,5)$	yes (T_2)	no	b
$(4,6)$	yes (T_2)	no	b
$(5,6)$	yes (T_2)	yes (P_2)	a

Tallying the column on the right:

$$a = 2, \qquad b = 4, \qquad c = 4, \qquad d = 5, \qquad a + b + c + d = 15 = \binom{6}{2}. \checkmark$$

5 The Rand Index

$$\mathrm{RI}(T,P) \;=\; \frac{a+d}{a+b+c+d} \;=\; \frac{a+d}{\binom{N}{2}}.$$

In words: the fraction of pairs on which T and P make the same statement. On our example,

$$\mathrm{RI} \;=\; \frac{2+5}{15} \;=\; \frac{7}{15} \;\approx\; 0.467.$$

Range: $[0,1]$. RI $= 1$ if and only if the two partitions are identical (every pair agrees); RI $= 0$ if every pair disagrees, which is rare and hard to construct.

So far, so good. The Rand Index is a clean, label-free way of measuring how close two partitions are. The problem is what value it gives in practice. We turn to that now.

6 Problem 1: random clusterings score embarrassingly high

Here is the first thing that should bother you: an utterly useless clustering does *not* score RI $= 0$. It scores something like 0.5, and the exact number depends on how the problem is set up.

A concrete experiment we can do by hand. Take a clean toy problem: $N = 30$ items, truth has two clusters of 15 each. Now I propose the dumbest possible prediction: independently for each item, flip a fair coin — heads goes into predicted cluster 1, tails into predicted cluster 2. The coin doesn't know anything about authorship. The prediction is pure noise.

What RI do we expect on average?

For any specific pair of items (i,j), the prediction puts them in the same cluster exactly when both coin flips agree (HH or TT), which happens with probability $\frac{1}{2}$. This is independent of whether truth puts them together — the coins do not consult the truth.

Counting the pairs.

- **True-same pairs**: items within the same true cluster. Each true cluster of 15 contributes $\binom{15}{2} = 105$ pairs, and there are two clusters, so $2 \cdot 105 = 210$ true-same pairs.
- **True-diff pairs**: every other pair. $\binom{30}{2} - 210 = 435 - 210 = 225$.

Expected agreements. For each true-same pair, the coin agrees (says "same") with probability $\frac{1}{2}$. So:

$$E[a] \;=\; 210 \cdot \tfrac{1}{2} \;=\; 105.$$

For each true-diff pair, the coin agrees (says "diff") with probability $\frac{1}{2}$. So:

$$E[d] \;=\; 225 \cdot \tfrac{1}{2} \;=\; 112.5.$$

And therefore:

$$E[\mathrm{RI}] \;=\; \frac{E[a]+E[d]}{\binom{30}{2}} \;=\; \frac{105+112.5}{435} \;=\; 0.500.$$

So a coin-flip clustering scores RI ≈ 0.5. It learned nothing, but half the pairs agreed with truth by accident. This means RI = 0.5 is *not* the midpoint between worst and best. It is the baseline. You would need RI well above 0.5 before claiming you have learned anything from the data.

This is annoying but not yet fatal. We will see the killer problem next.

7 Problem 2: the random baseline depends on K

Here is the deal-breaker. The number 0.5 we just computed is the baseline only for $K = 2$. It depends sensitively on how many clusters are in the truth. As K grows, the baseline grows toward 1, and RI loses its ability to discriminate.

The general setup. Suppose truth has K clusters of equal size M each, so $N = KM$. The prediction is again pure noise: each item dropped uniformly at random into one of K predicted clusters. For any pair of items,

$$P(\text{pred says same}) \;=\; \frac{1}{K}, \qquad P(\text{pred says diff}) \;=\; \frac{K-1}{K}.$$

(Because both items must land in the same one of K predicted clusters, which happens with probability $1/K$.)

Counting the true-same and true-diff pairs. True-same pairs: $K \cdot \binom{M}{2}$. True-diff pairs: $\binom{N}{2} - K\binom{M}{2}$.

Expected agreements.

$$E[a] \;=\; K\binom{M}{2} \cdot \frac{1}{K} \;=\; \binom{M}{2},$$

$$E[d] \;=\; \left[\binom{N}{2} - K\binom{M}{2}\right] \cdot \frac{K-1}{K},$$

$$E[\text{RI}] \;=\; \frac{E[a] + E[d]}{\binom{N}{2}}.$$

Now I plug in our project's setting — $M = 15$ articles per author — and sweep K. Each row of the table below is the result of one full calculation using the formulas above. Try one row yourself to verify; the arithmetic takes 30 seconds.

K	$N = KM$	$\binom{N}{2}$	true-same	true-diff	$E[a]$	$E[d]$
2	30	435	210	225	105	112.5
4	60	1,770	420	1,350	105	1,012.5
6	90	4,005	630	3,375	105	2,812.5
8	120	7,140	840	6,300	105	5,512.5
10	150	11,175	1,050	10,125	105	9,112.5
12	180	16,110	1,260	14,850	105	13,612.5

Table 1: Components of $E[\text{RI}]$ under a random K-cluster prediction, with $M = 15$ items per true cluster. Note that $E[a]$ stays constant at $\binom{M}{2} = 105$ while $E[d]$ explodes.

Dividing $E[a] + E[d]$ by $\binom{N}{2}$ gives the random baseline:

K	$E[\text{RI}]$ for random	what RI ≈ 0.85 would mean
2	0.500	far above chance: clustering is genuinely working
4	0.631	well above chance
6	0.729	clearly above chance
8	0.787	somewhat above chance
10	0.825	barely above chance
12	0.852	*at* chance: a fair coin would do this

Table 2: The same RI score means wildly different things at different K.

A single RI value, like 0.85, can simultaneously mean "great performance" (at $K = 2$) and "literally random" (at $K = 12$). If you plot RI vs K for a clustering algorithm of *constant skill*, RI will trend *upward* purely because the random baseline trends upward. The actual quality could be flat, climbing, or collapsing, and you cannot tell which from RI alone.

Why does the baseline climb with K? Look at the table. At $K = 12$ there are $14{,}850$ true-diff pairs out of $16{,}110$ total — that's 92%. A random prediction also produces many small predicted clusters, so the vast majority of its pairs are pred-diff too. The two graders agree on "these two items are different" *trivially*, because almost everything is different from almost everything. That flood of trivial d-agreements inflates RI.

At $K = 2$, by contrast, "true-diff" is only about half of pairs, so random doesn't get easy points for free.

The lesson: **RI rewards a clustering not for what it gets right, but for how lopsided the partition is**. Lopsided structures automatically have high RI baselines.

8 The fix, conceptually

We want to throw away the part of RI that comes from chance. Conceptually, once we know the random baseline $E[\text{RI}]$:

- Anything below $E[\text{RI}]$ is worse than random and should score negative.
- Exactly $E[\text{RI}]$ should score zero.
- $\text{RI} = 1$ should score one.

The simplest way to achieve this is a linear rescaling: subtract the baseline, then divide by the distance from the baseline to the maximum.

$$\text{ARI} \;=\; \frac{\text{RI} - E[\text{RI}]}{1 - E[\text{RI}]} \;=\; \frac{\text{how much better than chance}}{\text{how much better perfect would be than chance}}.$$

In words: *ARI is the fraction of the gap from random to perfect that your clustering closed.*

That sentence is the whole conceptual picture. The actual H–A formula people write down differs slightly in technical details, but it answers exactly this question. We need two more ingredients before we can write it cleanly:

1. A precise definition of "random", so we can compute $E[\text{RI}]$.
2. A small algebraic rearrangement that makes the formula prettier and more numerically stable.

9 What does "random" mean? The Hubert–Arabie null

In Section 6 we used a very informal random model: "flip a coin for each item". That worked for the calculation but leaves a fundamental choice unspecified: does our random predictor get to know the true cluster sizes? Does it get to pick its own cluster sizes? What if those choices accidentally match the truth's cluster sizes?

The standard answer, and the one ARI uses, is the **Hubert–Arabie null model** (1985). It is the natural choice and we will see why.

The null model. Take both partitions' cluster sizes as fixed.

- Truth has clusters of sizes $\alpha_1, \alpha_2, \ldots, \alpha_K$ — these are observed and we hold them constant.
- Pred has clusters of sizes $\beta_1, \beta_2, \ldots, \beta_L$ — also observed and held constant.

Under the null, the only thing that is random is *which items get which combination of (true, pred) labels*, subject to the row and column-size constraints.

Equivalently: build a contingency table whose row sums are (α_i) and column sums are (β_j), and fill in the cells uniformly at random subject to those marginals.

Why this null? Three reasons.

1. **It doesn't give you free credit for matching cluster sizes by accident.** The marginals are pinned to what we observed. Only the cross-tabulation is random.
2. **It is parameter-free.** No tuning constants. Both α_i and β_j are data.
3. **It is a classical model in statistics** — the same null behind Fisher's exact test. The math is well-understood.

10 The expected a under the null

To compute the expectation, we need to think about a in terms of the contingency table.

Re-deriving a from the table. Let n_{ij} be the count in cell (i, j) — the number of items that are in true cluster i and predicted cluster j. Two items *both* land in the same true cluster and the same predicted cluster exactly when they sit in the same cell. The number of pairs inside cell (i, j) is $\binom{n_{ij}}{2}$, so:

$$a = \sum_{i,j} \binom{n_{ij}}{2}.$$

This is just a rewrite of the definition of a, but it's the rewrite we need: now a is a sum over cells, and each cell's contribution depends only on a count n_{ij} whose distribution we know.

The distribution of n_{ij} under the null. Under the Hubert–Arabie null, the row marginals are α_i and the column marginals are β_j. The marginal distribution of n_{ij} is hypergeometric, with mean $E[n_{ij}] = \alpha_i \beta_j / N$ (natural: row i has α_i items, each independently sitting in column j with probability β_j/N).

A standard identity for the hypergeometric distribution gives the second moment we actually need:

$$E\left[\binom{n_{ij}}{2}\right] = \frac{\binom{\alpha_i}{2}\binom{\beta_j}{2}}{\binom{N}{2}}.$$

You can prove this by writing $\binom{n_{ij}}{2}$ as the number of pairs of items both landing in cell (i, j), and computing the probability for each pair separately — I will skip the proof here, but the formula is classical and you can take it on faith.

Summing over all cells.

$$
\begin{aligned}
E[a] &= \sum_{i,j} E\left[\binom{n_{ij}}{2}\right] = \sum_{i,j} \frac{\binom{\alpha_i}{2}\binom{\beta_j}{2}}{\binom{N}{2}} \\
&= \frac{1}{\binom{N}{2}} \left(\sum_i \binom{\alpha_i}{2}\right)\left(\sum_j \binom{\beta_j}{2}\right)
\end{aligned}
$$

where in the last step we used $\sum_{ij} x_i y_j = (\sum_i x_i)(\sum_j y_j)$.

Two clean quantities to name. Each of those sums has a meaning of its own.

$$
\begin{aligned}
S_T &= \sum_i \binom{\alpha_i}{2} = \text{(number of true-same pairs)}. \\
S_P &= \sum_j \binom{\beta_j}{2} = \text{(number of pred-same pairs)}.
\end{aligned}
$$

Then

$$
E[a] = \frac{S_T \cdot S_P}{\binom{N}{2}}.
$$

That's it. S_T is a property of truth alone; S_P is a property of prediction alone; their product divided by $\binom{N}{2}$ is the chance-level number of TS–PS pairs we'd expect if there were no relationship between them.

11 The actual ARI formula

Recall the conceptual ARI:

$$
\text{ARI} = \frac{\text{observed agreement above chance}}{\text{maximum possible agreement above chance}}.
$$

Hubert and Arabie wrote this using a directly rather than RI, which gives an equivalent but cleaner formula:

$$
\text{ARI} = \frac{a - E[a]}{\text{MaxIndex} - E[a]},
$$

where

$$
E[a] = \frac{S_T \cdot S_P}{\binom{N}{2}}, \qquad \text{MaxIndex} = \frac{S_T + S_P}{2}.
$$

Why is using a equivalent to using RI? A useful identity: among the four counts (a, b, c, d), only one is independent once the marginals are fixed. Specifically, $a + b = S_T$ (every true-same pair is either TS–PS or TS–PD), so $b = S_T - a$. Similarly $a + c = S_P$, so $c = S_P - a$. And $d = \binom{N}{2} - a - b - c$. This means b, c, d are all linear functions of a (once marginals are fixed), and so is RI. Adjusting a for chance and adjusting RI for chance give equivalent formulas, differing only by an overall scale that cancels in the ratio.

Why MaxIndex $= (S_T + S_P)/2$? The largest a can ever be is $\min(S_T, S_P)$ — you can't have more TS–PS pairs than the total of either type. So the *exact* maximum is $\min(S_T, S_P)$, and you could in principle put that in the denominator. H–A use the average $(S_T + S_P)/2$ instead.

Why? It is for algebraic convenience — the formula derived from $E[\binom{n_{ij}}{2}]$ matches the H–A denominator exactly, so the whole formula can be written cleanly in terms of S_T and S_P. With the average:

- ARI $= 1$ iff T and P are identical partitions (so $S_T = S_P = a$, and both numerator and denominator equal $S_T - E[a]$).
- ARI can be slightly negative for worse-than-random clusterings, which is informative and not a bug.
- ARI is symmetric in T and P, as it should be.

12 Putting it all together: ARI for our running example

Back to the $N = 6$ example from Section 4: $T = \{\{1, 2, 3\}, \{4, 5, 6\}\}$, $P = \{\{1, 2, 4\}, \{3, 5, 6\}\}$.

Step 1: build the contingency table. For each item we look at its (T, P) pair of labels and tally.

	$P_1 = \{1, 2, 4\}$	$P_2 = \{3, 5, 6\}$	row total α_i
$T_1 = \{1, 2, 3\}$	2 (items 1, 2)	1 (item 3)	3
$T_2 = \{4, 5, 6\}$	1 (item 4)	2 (items 5, 6)	3
col total β_j	3	3	$6 = N$

Step 2: read off a from the table.

$$a \;=\; \sum_{i,j} \binom{n_{ij}}{2} \;=\; \binom{2}{2} + \binom{1}{2} + \binom{1}{2} + \binom{2}{2} \;=\; 1 + 0 + 0 + 1 \;=\; 2.$$

(Matches what we counted by hand in Section 4.)

Step 3: compute S_T and S_P.

$$S_T \;=\; \binom{3}{2} + \binom{3}{2} \;=\; 3 + 3 \;=\; 6.$$

$$S_P \;=\; \binom{3}{2} + \binom{3}{2} \;=\; 3 + 3 \;=\; 6.$$

Step 4: compute $E[a]$.

$$E[a] \;=\; \frac{S_T \cdot S_P}{\binom{N}{2}} \;=\; \frac{6 \cdot 6}{15} \;=\; \frac{36}{15} \;=\; 2.4.$$

Interpretation: with these marginals (two clusters of three on each side), a random arrangement would, on average, produce 2.4 TS–PS pairs. We observed 2. So we're slightly *below* expectation.

Step 5: compute MaxIndex and ARI.

$$\text{MaxIndex} \;=\; \frac{S_T + S_P}{2} \;=\; \frac{6 + 6}{2} \;=\; 6.$$

$$\text{ARI} \;=\; \frac{a - E[a]}{\text{MaxIndex} - E[a]} \;=\; \frac{2 - 2.4}{6 - 2.4} \;=\; \frac{-0.4}{3.6} \;\approx\; -0.111.$$

Sanity-checking the negative value. RI for this example was 0.467, which sounds OK. But ARI says we are *worse* than chance. Both can be true: under the H–A null with these specific marginals (two clusters of three on each side), the average random arrangement actually produces 2.4 TS–PS pairs — more than the 2 we observed. So our particular swap (item 3 and item 4) does worse than typical noise.

This is the value of ARI: it tells you not just "how many pairs match" but "how many pairs match *compared to what nothing-going-on would have produced*".

13 Three more sanity checks

To make sure the formula is doing what we want, let's run it on three quick examples (still $N = 6$, same truth T as before).

13.1 Perfect prediction

$P = T = \{\{1, 2, 3\}, \{4, 5, 6\}\}$. Contingency table: $\left[\begin{smallmatrix} 3 & 0 \\ 0 & 3 \end{smallmatrix}\right]$.

$$a = \binom{3}{2} + \binom{3}{2} = 6, \quad S_T = 6, \quad S_P = 6, \quad E[a] = 2.4, \quad \text{MaxIndex} = 6.$$

$$\text{ARI} = \frac{6 - 2.4}{6 - 2.4} = 1. \quad \checkmark$$

13.2 All in one cluster

$P = \{\{1, 2, 3, 4, 5, 6\}\}$. Contingency table: $\left[\begin{smallmatrix} 3 \\ 3 \end{smallmatrix}\right]$ (one column).

$$a = \binom{3}{2} + \binom{3}{2} = 6, \quad S_T = 6, \quad S_P = \binom{6}{2} = 15, \quad E[a] = \tfrac{6 \cdot 15}{15} = 6, \quad \text{MaxIndex} = \tfrac{6+15}{2} = 10.5.$$

$$\text{ARI} = \frac{6 - 6}{10.5 - 6} = 0. \quad \checkmark$$

Putting everything in one cluster gets the same agreement as random — because it has *no information* about cluster structure at all.

13.3 Each item in its own cluster

$P = \{\{1\}, \{2\}, \{3\}, \{4\}, \{5\}, \{6\}\}$. Contingency table: each row has three columns with 1 in three of the six.

$$a = 6 \cdot \binom{1}{2} = 0, \quad S_T = 6, \quad S_P = 6 \cdot \binom{1}{2} = 0, \quad E[a] = \tfrac{6 \cdot 0}{15} = 0, \quad \text{MaxIndex} = 3.$$

$$\text{ARI} = \frac{0 - 0}{3 - 0} = 0. \quad \checkmark$$

Also gets $\text{ARI} = 0$. Singletons-everywhere is exactly as informative as all-in-one — you can extract no structure from either.

13.4 A reasonable mistake

$P = \{\{1,2\},\{3,4,5,6\}\}$ — we picked up the first true cluster mostly right but lost item 3 into the merged blob with T_2.

Contingency:

	$P_1 = \{1,2\}$	$P_2 = \{3,4,5,6\}$
$T_1 = \{1,2,3\}$	2	1
$T_2 = \{4,5,6\}$	0	3

$$a = \binom{2}{2} + \binom{1}{2} + \binom{0}{2} + \binom{3}{2} = 1 + 0 + 0 + 3 = 4.$$

$$S_T = 6, \quad S_P = \binom{2}{2} + \binom{4}{2} = 1 + 6 = 7.$$

$$E[a] = \tfrac{6\cdot 7}{15} = 2.8, \quad \text{MaxIndex} = \tfrac{6+7}{2} = 6.5.$$

$$\text{ARI} = \frac{4 - 2.8}{6.5 - 2.8} = \frac{1.2}{3.7} \approx 0.324.$$

A clearly-better-than-random but imperfect prediction. The intermediate ARI value reflects that. Note also that RI for this case is $(4+4)/15 \approx 0.53$ — only modestly above the random baseline. ARI translates that modest excess into a more interpretable number.

14 Back to your K-sweep

Recall the question that started this document: why is ARI the right metric for the K-sweep, instead of RI?

Section 7 answered the conceptual half. Now we can give the quantitative answer: **ARI of a random prediction is** 0 **at every** K, by construction. The numerator $a - E[a]$ has expectation zero under the null, so the ratio does too. The chance baseline that ballooned RI from 0.5 to 0.85 across the K-sweep — entirely parasitic, entirely uninformative — gets subtracted out completely by ARI.

This means: when we plot ARI vs K for our K-means runs, every point on the plot is measured against the same zero. A drop from ARI $= 0.65$ at $K = 2$ to ARI $= 0.31$ at $K = 12$ tells us the same story we'd see on a fair scale: clustering quality is genuinely degrading as the author pool grows. RI would have shown the opposite.

To make this very concrete, here are the K-sweep results from our LessWrong 35 Authors experiment, alongside the approximate RI a random prediction would score at each K (from Table 2).

K	ARI (K-means)	random's RI	approx K-means RI
2	0.65	0.500	~ 0.83
4	0.54	0.631	~ 0.83
6	0.47	0.729	~ 0.86
8	0.38	0.787	~ 0.87
10	0.36	0.825	~ 0.89
12	0.31	0.852	~ 0.90

Table 3: ARI tells the truth (clustering gets harder); RI tells the opposite (looks like it gets easier), purely because the random baseline grows. The K-means RI is approximate because it depends on actual cluster sizes in each trial, but the qualitative trend is robust.

If we had naively reported RI for this experiment, we would have written "RI improves from 0.83 to 0.90 as we add more authors" — a completely backwards conclusion that would have invalidated half of the report.

15 What to remember

1. Comparing two partitions = counting how many *pairs of items* the partitions agree on (same-or-different).
2. Pairs fall into four buckets: a, b, c, d. RI = $(a + d)/\binom{N}{2}$.
3. A random clustering scores RI that is *not* zero, and *not constant* across different cluster counts K. The baseline rises with K because the universe of pairs becomes dominated by "trivially different".
4. ARI subtracts the random baseline. Conceptually: ARI = (RI $-$ E[RI]) / (1 $-$ E[RI]). Practically, the Hubert–Arabie form is

$$\text{ARI} \;=\; \frac{a - E[a]}{\text{MaxIndex} - E[a]}, \qquad E[a] = \frac{S_T S_P}{\binom{N}{2}}, \qquad \text{MaxIndex} = \frac{S_T + S_P}{2}.$$

5. By construction, ARI $= 0$ for random, ARI $= 1$ for perfect, ARI < 0 for worse than random. These values mean the same thing across different K, different cluster sizes, different datasets — which is the whole reason we use it.